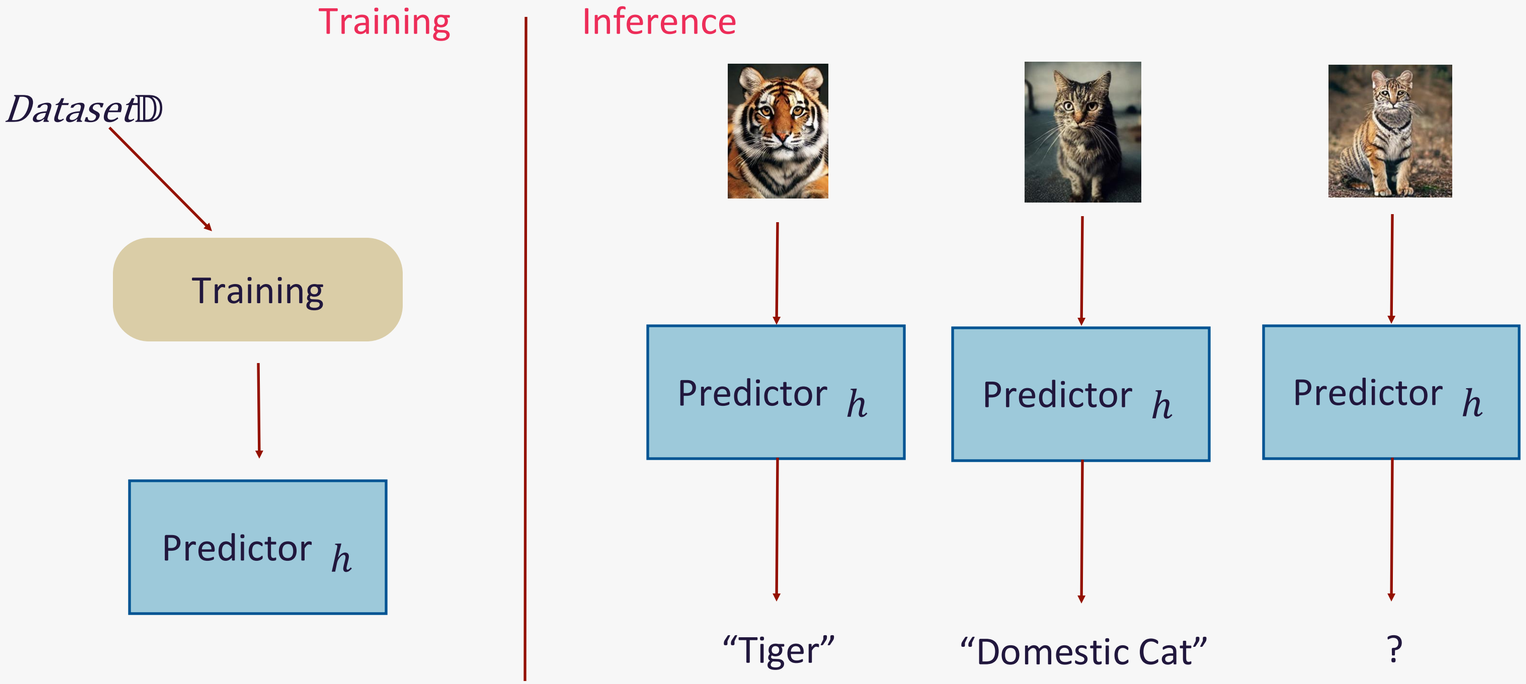
Explainable artificial intelligence

Instructor: Giorgos Stamou

Providing accurate explanations

images created by DALL·E

Deep learning systems operate by identifying statistical complex patterns in data, rather than applying explicit, symbolic reasoning



Key question

[Context: A quick glance at an image is often enough for a human to identify features or patterns of interest and label an animal image as"tiger"]

Is it possible to explain how does h assign labels?

Context: tigers and domestic cats

Tigers

A tiger is a large, powerful wild cat known for its distinctive orange coat with black stripes. Here are some key physical characteristics:

- Size: Tigers are the largest of all wild cats, with males reaching up to 3.3 meters (11 feet) in length, including their tails.
- Fur: Their fur is usually bright orange with black stripes that run vertically along their bodies.
- White Markings: They have white fur on the belly, inner legs, cheeks, and around the eyes.
- Tail: Long and striped, often with a white tip.
- Eyes: Large, golden-yellow or amber-colored eyes with round pupils.

Domestic cats

A domestic cat is a small, agile, and furry feline that comes in a variety of breeds, colors, and patterns. Here are the key features of a typical domestic cat:

- Size: Small, typically weighing between 2 to 7 kg (5 to 15 lbs).
- Body Shape: Lean, muscular, and agile, with a graceful and flexible body.
- Fur: Short or long, depending on the breed, with various patterns and colors.
- Tail: Long and expressive, sometimes fluffy (depending on breed).
- Eyes: Large, round or almond-shaped, with colors ranging from blue, green, yellow, hazel, or even odd-colored (heterochromia).

descriptions created by chatGPT; image created by DALL·E

Context: tigers and domestic cats

Feature	Tiger	Domestic Cat
Size & Body Structure	Large, muscular build	Small, lightweight, and more delicate
Fur Pattern	Bold, thick black stripes on an orange coat	Fainter, thinner stripes (tabby-like)
Facial Features	Broad face with a strong jaw, smaller rounded ears	Smaller face, larger pointed ears
Tail	Thick and short	Long and thin
Paws & Claws	Large paws with strong retractable claws	Small paws with softer, retractable claws

descriptions created by chatGPT;
image created by DALL·E

- “Instance A was classified as a tiger because it has bold black stripes on an orange coat, a large, muscular body, and a broad facial structure with small rounded ears. These features strongly match the tiger images in my training dataset.”
- “Instance B was classified as a domestic cat because it has a much smaller body size, a thinner build, and rounder eyes. While it has striped fur, the stripes are narrow and irregular, more similar to a cat than a tiger. Additionally, the tail is longer and thinner, whereas tigers have thick, powerful tails.”
- “Instance A was classified as a tiger, while Instance B was classified as a domestic cat, even though they look similar. This is likely because Instance A was photographed in a jungle setting, which might have influenced the model, whereas Instance B was photographed indoors, making it appear more like a domestic animal.”

Which is the most effective context to describe the functioning of the predictor?

Context: tigers and domestic cats

Feature	Tiger	Domestic Cat
Size & Body Structure	Large, muscular build	Small, lightweight, and more delicate
Fur Pattern	Bold, thick black stripes on an orange coat	Fainter, thinner stripes (tabby-like)
Facial Features	Broad face with a strong jaw, smaller rounded ears	Smaller face, larger pointed ears
Tail	Thick and short	Long and thin
Paws & Claws	Large paws with strong retractable claws	Small paws with softer, retractable claws

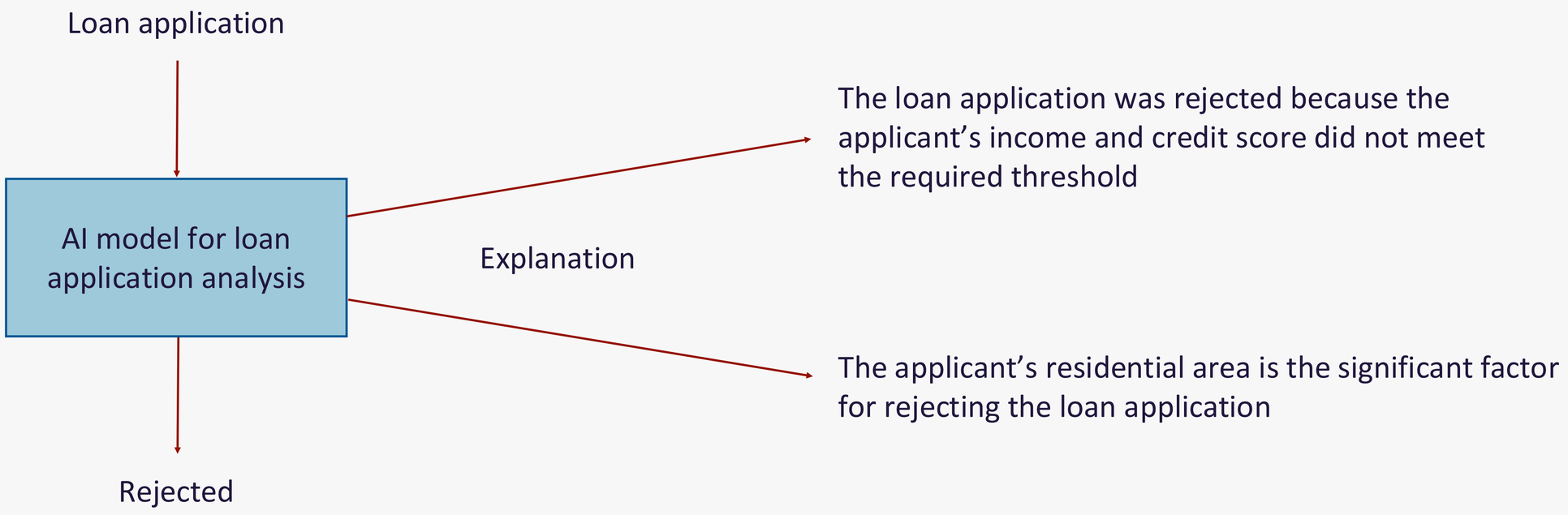
descriptions created by chatGPT; image created by DALL·E

- Does the human brain distinguish between tigers and domestic cats based on these physical differences?
  - The human brain employs a multifaceted process involving visual processing, memory, experience, and cognitive reasoning to recognize objects and entities
  - AI predictors like deep neural networks may follow intricate processes to predict

Explanations in specific contexts may offer intuitive descriptions of the differences in the appearance of tigers and cats; but, these explanations might be misleading for the functioning of the AI system: they may not be informative for the process of the AI predictor

Explanations should emphasize the key factors that led to the prediction or decision

Approximate the logic of deep neural networks in human-understandable terms

Deep neural networks, excel at detecting statistical patterns within datasets

- If the training data contains bias, the model may learn and reinforce these biases
- Explanations should not obscure this issue but instead highlight it